

SmartCare v0.4 - Domain Implementation Workbook

Week 7 student resource

1. UML-to-Code Trace

UML element	Python element	Implemented?	Notes
Patient (class)	Class patient	Yes	
Patient.patient_name	Self._patient_name (private attribute) with get_patient_name()/set_patient_name()	Yes	
Patient.add_patient()	Patient.__init__()	Yes	
Patient.find_patient()	_	No	
Practitioner (class)	Class practitioner	Yes	
Practitioner.practitioner_name	Self._practitioner_name (private attribute) with get_practitioner_name()	Yes	
Practitioner.add_practitioner()	Practitioner.__init__()	Yes	
Practitioner.view_appointments()		No	
Appointment (class)	Class Appointment	Yes	
Appointment.appointment_time/status	Self._appointment_time, self._status (private attributes) with get_methods	Yes	
Appointment.create()/update()/cancel()	__init__(), set_appointment_time(), cancel(), complete()	Yes	

2. Domain Invariants

Class	Invariant / rule	How protected
Patient	Patient_name can not be empty	Checked with an if/else
Practitioner	Practitioner_name can never be empty	Same as patient
Appointment	Status may only move SCHEDULED – COMPLETED or SCHEDULED - CANCELLED	Not use if/else that only allows the change while status is still scheduled
Appointment	A cancelled appointment is never deleted, only marked cancelled	No delete/remove method exists on appointment
Appointment	Every appointment references exactly one patient and one practitioner	No way to create an appointment without putting them into both patient and practitioner

3. Composition / Inheritance Decisions

Relationship	Decision	Rationale
Appointment - Patient	Association	Appointment stores a reference to a patient object

Appointment - Practitioner	Association	Same as patient. A practitioner can be booked for one appointment
Doctor – Practitioner	Inheritance	Hypothetical
Clinic - Appointment	Not at this stage	Not confirmed

4. AI Pair-Programming Record

AI contribution	Conforms?	Decision	Reason	Verification
__init__ setting __status to "SCHEDULED" automatically	Yes	Kept	Matches the UML status	get_status() returns "SCHEDULED" right after creating an Appointment
cancel()/complete() checking current status with if/else before changing	Yes	Kept	Uses only if/else and return values	
Initial draft included an unchecked set_status() method	No	Rejected	Any code can overwrite	It only has a cancel and complete
Initial draft imported a NotificationManager	No	Rejected	Out of Scope	Not needed
Initial draft had Appointment inherit from a generic Record class	No	Rejected		

5. Updated UML

Insert updated UML only if implementation revealed a justified design change. Explain every change.

No changes were made to the diagram however my one from last week uploaded half cut of so was reuploaded at the top to show the whole diagram. It is still the same three classes, patient, practitioner and appointment.